

👉 XAI: Chess Knowledge in AlphaZero & Interpreting Stable Diffusion

FAMAF - UNC

First Semester 2026



Disclaimer ---

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

Spring 2023, Harvard University

<https://interpretable-ml-class.github.io/>

Acquisition of Chess Knowledge in AlphaZero

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¹DeepMind

²Google Brain

³World Chess Champion, 2000–2007*

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ABSTRACT

What is learned by sophisticated neural network agents such as AlphaZero? This question is of both scientific and practical interest. If the representations of strong neural networks bear no resemblance to human concepts, our ability to understand faithful explanations of their decisions will be restricted, ultimately limiting what we can achieve with neural network interpretability. In this work we provide evidence that human knowledge is acquired by the AlphaZero neural network as it trains on the game of chess. By probing for a broad range of human chess concepts we show when and where these concepts are represented in the AlphaZero network. We also provide a behavioural analysis focusing on opening play, including qualitative analysis from chess Grandmaster Vladimir Kramnik. Finally, we carry out a preliminary investigation looking at the low-level details of AlphaZero's representations, and make the resulting behavioural and representational analyses available online.

1 Introduction

(McGrath et al. 2022)

🍷 Learning from machines (AlphaZero) ---

- Most methods we have looked so far try to interpret algorithms trained on human-generated data and labels
 - ▶ These interpretations may resemble human-understandable representations only because they learned from such data
- Can we interpret what the algorithms has been learning through its self-play training process?



🍷 Learning from Machines: Three-Pronged Acquisition ---

- **Probe for concepts**

- ▶ How closely is AlphaZero's internal representation **related to** chess concepts humans have already created?
- ▶ Detection of human concepts from network activations

- **Study behavioral changes**

- ▶ How do changing representations give rise to changing behaviors?
- ▶ Evolution of AlphaZero vs. human's strategy in openings

- **Investigate activations directly**

- ▶ Unsupervised methods using non-negative matrix factorization (NMF) and direct measure of covariance to discover concepts not represented by humans in the past

Interpretability ---

Concept-based (post-hoc) interpretability

- **Network probing:** detect human concepts from internal activations using linear classifiers (*concept activation vectors*)
- **Important features:** which concepts matter most for model predictions?
- **Mechanistic understanding:** what algorithms do individual layers implement?
- **Key challenge:** probing measures *correlation*, not *causation* — a concept linearly decodable from activations may not causally drive behavior

Explainability in reinforcement learning

- **Structural causal models:** simulate interventions on actions to answer counterfactual questions (*what if a different move was played?*)
- **Reward difference explanations:** why was action a chosen over a' ? Quantify the expected reward gap
- **Behavioral trajectory analysis:** identify critical decision points where the agent's policy changes most dramatically

🍷 Chess as Testing Ground for AI Interpretability ---

Can humans learn the machine's strategy?

- Does not rely on human-labeled data
- Tree-based organization / saliency maps
- Natural language processing to generate move-by-move commentary
- **This paper:** captures “intuitive” aspect of chess play by understanding networks that produce value assessment (**v**) and candidate move (**p**)

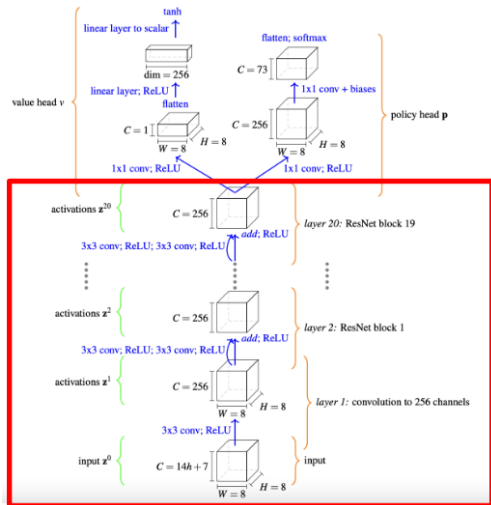
$$\mathbf{p}, v = f_{\theta}(\mathbf{z}^0)$$

In plain English

$\mathbf{z}^0$ is the board position fed to the network; $\mathbf{p}$ is a probability distribution over legal moves (what AlphaZero *wants* to play), and $v \in [-1, 1]$ is its estimate of who is winning. The question of the paper is: *what does f_{θ} know about chess to produce these outputs?*



AlphaZero: Network Structure and Training ---



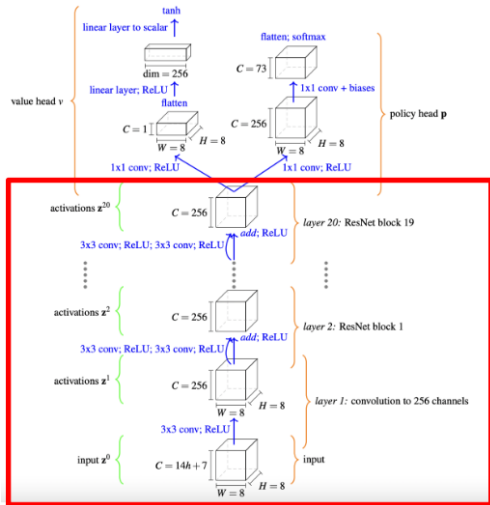
$$\mathbf{p}, v = f_{\theta}(\mathbf{z}^0)$$

$$\mathbf{z}^l = f^l(\mathbf{z}^{l-1}) = \text{ReLU}(\mathbf{z}^{l-1} + g^l(\mathbf{z}^{l-1}))$$

$$\mathbf{z}^l = f^{1:l}(\mathbf{z}^0) = f^l \circ \dots \circ f^2 \circ f^1(\mathbf{z}^0)$$



AlphaZero: Network Structure and Training ---



In plain English

In the diagram, each box in the red rectangle is one residual block. The **“add; ReLU”** arrow inside each box is where the network adds the unmodified input to the learned transformation — instead of replacing it. This repeats 20 times, building progressively more abstract representations of the board.

Probing for Concepts - Encoding of Human Conceptual Knowledge ---

Question: Can human concepts be easily predicted from the network's internal representation?

Concept: User-defined function mapping network input to real line:

$$c(\mathbf{z}^0) = \begin{cases} 1 & \text{if } \mathbf{z}^0 \text{ contains a bishop-pair U+2657 for the playing side} \\ 0 & \text{otherwise} \end{cases}$$

Approach: Train a sparse linear regression model from activations $\mathbf{z}^l$ at layer l and training step t to human concept j

In plain English

A *concept* is any chess idea a human can define on a board position — from simple material facts (does the player have the bishop pair?) to complex positional ideas (is the king safe?). If a simple linear model can predict c_j from $\mathbf{z}^l$, the concept is **linearly encoded** in

Key Insight

The network was never explicitly trained to represent chess concepts — it only learned to predict moves and outcomes via self-play. Finding linearly decodable concepts means AlphaZero **spontaneously developed human-like internal representations**.

Probing Concept Learning: Details ---

- **Data:** Randomly sample training, validation, test data from ChessBase archive, compute concept values and AlphaZero activations
- **Procedure:** For concept i , layer l , training step t , solve:

$$\mathbf{w}_{jlt}, b_{jlt} = \min_{\mathbf{w}, b} \frac{1}{N} \left\| \mathbf{w}^T \mathbf{z}_t^l + b \mathbf{1} - \mathbf{c}_j \right\|_2^2 + \lambda \|\mathbf{w}\|_1 + \lambda |b|$$

- **Controls:** Regression from $\mathbf{z}^0$ and random concept regression
- **Evaluation:** R^2 value, fraction of variance in concept explained by network activation

In plain English

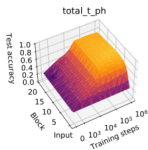
Fit a Lasso regression: predict concept c_j from activations $\mathbf{z}^l$. The ℓ_1 penalty keeps only the **few neurons** that actually encode the concept.

Key Insight

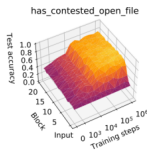
High R^2 = the concept is **linearly readable** from that layer. Repeat for every layer l and checkpoint $t \rightarrow$ a map of *what, when, and where* each concept is learned.



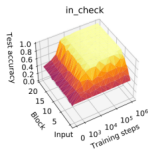
Evolution of Human Concepts in AlphaZero 1/2 ---



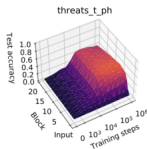
(a) Stockfish 8's total score



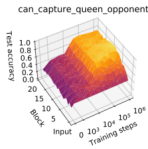
(b) A contested open file is occupied by rooks and/or queens of opposite colours



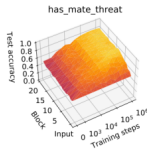
(c) Is the playing side in check?



(d) Stockfish 8's evaluation of threats.



(e) Can the playing side capture their opponent's queen?



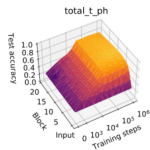
(f) Could the opposing side checkmate the playing side in one move?

Each surface is a *what-when-where* plot: probe R^2 as a function of **network depth** (Block) and **training time** (Training steps).

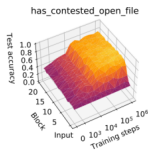
- **x-axis:** training steps ($10^0 \rightarrow 10^6$) — *when* is it learned?
- **y-axis:** block index (1–20) — *where* in the network?
- **z-axis:** test accuracy — *how well* is the concept encoded?



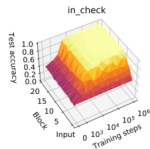
Evolution of Human Concepts in AlphaZero 2/2 ---



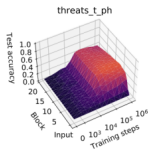
(a) Stockfish 8's total score



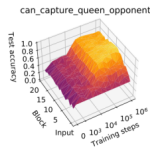
(b) A contested open file is occupied by rooks and/or queens of opposite colours



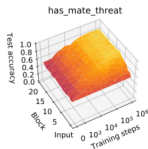
(c) Is the playing side in check?



(d) Stockfish 8's evaluation of threats.



(e) Can the playing side capture their opponent's queen?



(f) Could the opposing side checkmate the playing side in one move?

- A surface that rises early and stays high means AlphaZero learned that concept **fast and robustly** (e.g. `in_check`).
- A surface that stays flat means the concept is **never linearly encoded** (e.g. `threats_t_ph` — still dark/purple throughout).
- Simple tactical concepts (`in_check`, `can_capture_queen`) emerge **early and in middle layers**. Complex strategic concepts (`total_t_ph`, `has_mate_threat`) require **more training steps** and tend to concentrate in **deeper layers** — mirroring how humans learn chess: tactics before strategy.

Key Findings -- ---

- ① **Grokking:** Most concepts emerge abruptly around **32,000 training steps** accuracy is near zero before, then jumps and stabilizes. AlphaZero does not learn gradually; it *clicks*.
- ② **Information drop in deep layers:** for some concepts, probe accuracy *decreases* in the last blocks — the network re-encodes information in a less linearly-separable form as it approaches the output heads.
- ③ **Highly-distributed representations:** some concepts **cannot** be probed — sparsity forces the probe to use few neurons, but if the concept is spread across many neurons simultaneously, a sparse linear probe cannot recover it.
- ④ **Learning from errors:** positions where the probe fails systematically may reflect a genuine **“difference of opinion”** between AlphaZero and Stockfish — not a probe failure, but AlphaZero evaluating the position by a different (possibly superior) criterion.

Key Insight

Findings 3 and 4 are honest about the method's limits: absence of a detectable concept $\neq$ absence of the concept — it may simply be encoded in a way a **linear probe cannot see**.

🍲 Challenges for Concept Probing ---

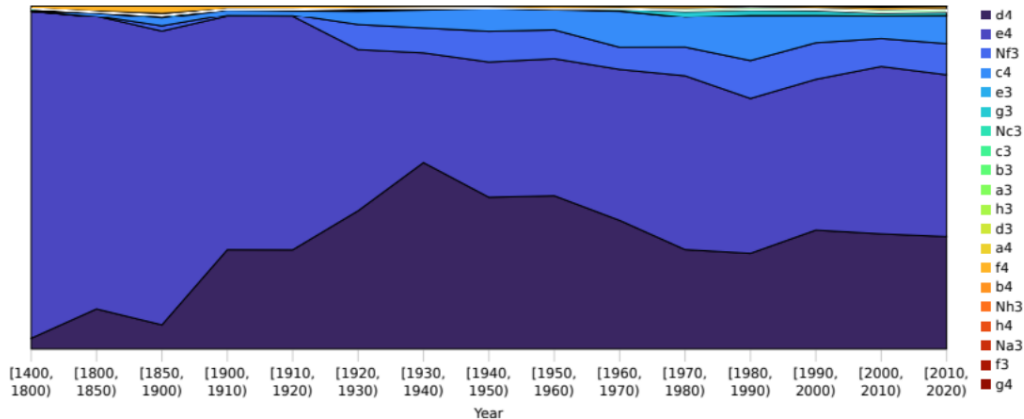
- ➊ What's the right **probing architecture**?
- ➋ How should we interpret **complex or subjective concepts**?
- ➌ When can we definitively say a **concept is represented**?
- ➍ When we train a probe, we cannot tell if we are getting a **confounder or the concept** itself

Progression through AlphaZero and Human History

Progression of Knowledge ---

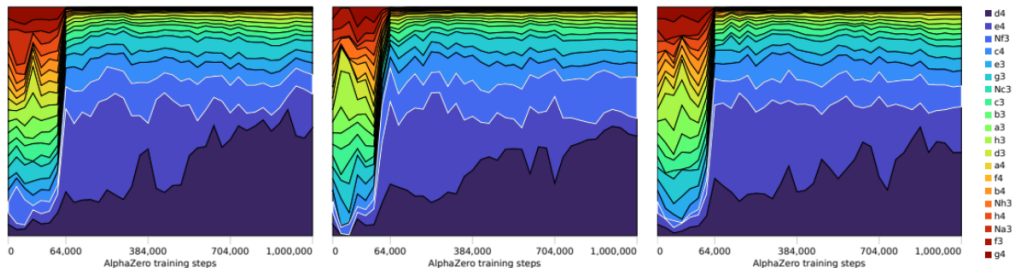
- Recall: $\mathbf{p}$, $v = f_{\theta}(\mathbf{z}^0)$
- **Question:** How does the progression of AlphaZero's knowledge compare to that of humans?
- **Key Findings:**
 - ▶ AlphaZero: Start from a uniform prior, then narrow down.
 - ▶ Humans: Start from a concentrated prior, then expand.

🍷 Progression through Human History ---



Each color represents a first move (e.g. e4, d4, Nf3), with its area showing how often humans played it across history — serving as a reference to compare whether AlphaZero's training recapitulates or diverges from human chess evolution.

🍷 Progression through AlphaZero History ---



The three panels show **three independent AlphaZero training runs**: the distribution is remarkably consistent across runs — AlphaZero converges to similar move preferences regardless of random initialization. Before 64,000 steps, chaotic exploration dominates; after that, the distribution stabilizes abruptly (**grokking**). Compared to human history, AlphaZero settles on a mix of **d4** and **e4** but with a more diverse and volatile distribution — **it does not recapitulate human chess history**, but finds its own path.

🍷 Progression of AlphaZero's Chess Knowledge ---

Methodology: At different training checkpoints (up to 128k steps), examine AlphaZero's move tendencies and concept probe accuracy.

Primary Takeaways

- ① *AlphaZero learns standard opening theory **early** — move preferences stabilize and match known theory within the first 64k steps*
- ② *AlphaZero learns **material values before positional concepts** — piece counts are encoded first; king safety and mobility come later*

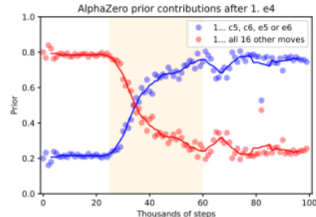
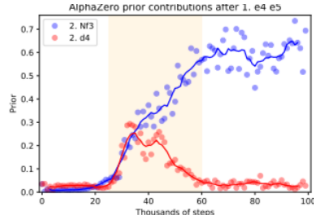
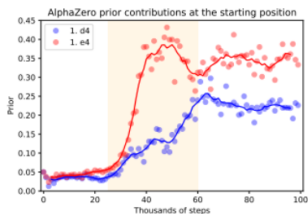
Both findings reinforce that AlphaZero learns basic human chess concepts first

Key Insight

This mirrors the curriculum of a human chess student: openings and material counting are taught before strategic positional play. AlphaZero rediscovers this ordering **from scratch**, via self-play alone — with no exposure to human games or pedagogy.

🍷 Opening Theory Knowledge ---

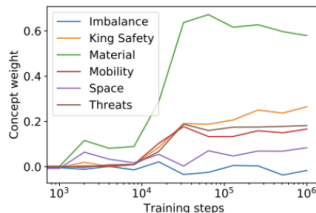
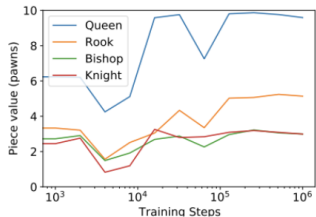
- 1 ~30–60k: AlphaZero plays 1. e4/d4 the majority of the time
- 2 ~45k: AlphaZero considers 2. d4 before opting for 2. Nf3
- 3 ~45k: AlphaZero plays standard responses to 1. e4



- **Graph 1:** What move should White play to open the game?
- **Graph 2:** What should White play on move 2, after Black responds with e5?
- **Graph 3:** How should Black respond to White's opening move e4?



Material vs. Positional Knowledge ---



Top — Material:

The *relative value of pieces*, measured in pawns. Standard human values are: Queen=9, Rook=5, Bishop=3, Knight=3. AlphaZero independently rediscovers approximately these values by $\sim 100k$ steps.

Bottom — Positional concepts:

- **Material:** piece count advantage
- **King Safety:** exposure to attacks
- **Mobility:** number of legal moves available
- **Space:** board control
- **Threats:** can capture next move
- **Imbalance:** asymmetric piece differences (e.g. bishop vs. knight)

The story they tell together: AlphaZero first learns to count pieces (material, $\sim 30k$ steps), and only later develops subtler positional concepts like king safety and mobility — exactly the order in which a human learns chess.

🍁 Training Progression Assessment: GM Vladimir Kramnik ---

Qualitative evaluation by World Chess Champion Vladimir Kramnik, who analyzed AlphaZero's play at three training checkpoints:

- ① **16k to 32k steps:** understands material value in complex positions — knows which pieces are worth sacrificing and which are not
- ② **32k to 64k steps:** develops king safety awareness in imbalanced positions — starts recognizing when the king is genuinely at risk
- ③ **64k to 128k steps:** combines king safety with material sacrifices in complex positions — a hallmark of grandmaster-level strategic play



Key Insight:

- Tactical skills appear to precede positional skills as AlphaZero learns
- This is a **human expert's qualitative confirmation** of what the concept probes showed quantitatively: AlphaZero's learning curriculum — material first, positional concepts later — mirrors how chess mastery develops in humans.

Exploring Additional Feature Detectors in AlphaZero Network

🍷 Exploring Activations with Unsupervised Methods ---

Goal: Find feature detectors embedded within the network — *without* using predefined human concepts as supervision.

In plain English

Unlike probing (which asks “*is concept X encoded here?*”), these methods ask “*what patterns exist here?*” with no human concept assumed in advance.

Key Insight

If unsupervised methods independently recover human-like concepts, it strengthens the case that AlphaZero’s representations are genuinely structured around chess knowledge — not an artifact of the linear probe design.

Unsupervised Methods: Details ---

- a) **Non-Negative Matrix Factorization (NMF):** decompose each layer's activations into a small set of additive factors — each factor represents a pattern of co-activating channels across board positions
- b) **Input-activation correlation:** measure the covariance between each channel's activation and the raw input board — reveals which input features drive individual neurons

Primary Takeaway

Individual layers and channels encode *feature detectors* related to human-recognizable chess concepts — discovered **without any concept labels**.

🍲 Approach #1: Non-Negative Matrix Factorization ---

Idea: compress each layer's activations into $K < C$ interpretable factors, then visualize each factor on the board to find *feature detectors*.

Step 1: Stack all activations for layer l with C channels: $\hat{\mathbf{Z}}^l \in \mathbb{R}^{NHW \times C}$

Step 2: Factorize $\hat{\mathbf{Z}}^l \approx \mathbf{\hat{z}}_{\text{all}} \mathbf{F}$:

$$\mathbf{F}^*, \mathbf{\hat{z}}_{\text{all}}^* = \min_{\mathbf{F}, \mathbf{\hat{z}}_{\text{all}}} \left\| \hat{\mathbf{Z}}^l - \mathbf{\hat{z}}_{\text{all}} \mathbf{F} \right\|_2^2 \quad \mathbf{F}, \mathbf{\hat{z}}_{\text{all}} \geq 0$$

- $\mathbf{\hat{z}}_{\text{all}} \in \mathbb{R}^{NHW \times K}$ — factor scores per board square
- $\mathbf{F} \in \mathbb{R}^{K \times C}$ — which channels compose each factor

Step 3: For each factor k and input n , visualize $\hat{z}_{nk}$ on the board.

Approach #1: Results ---

In plain English

Each factor k groups channels that fire together. Plotting its scores on the board reveals *where* that pattern activates — if it traces diagonals or controlled squares, the network learned that concept without being told to.

Key Insight

These patterns emerge with **no concept labels** — NMF discovers structure purely from activation co-occurrence. If the recovered factors resemble human chess concepts, it independently confirms the probing results.

🍷 Results: NN Matrix Factorization Analysis ---



Diagonal Moves
(1st Layer)



Diagonal Moves
(2nd Layer)



of Pieces that could
move to each square

Approach #2: Input-Activation Covariance Analysis ---

Idea: for each channel i in layer l , measure how strongly its activation correlates with each input feature — then visualize on the board.

Step 1: Compute covariance between channel i 's activation z_i^l and the raw input board $\mathbf{z}^0$:

$$\text{cov}(z_i^l, \mathbf{z}^0) = \mathbb{E} [z_i^l \mathbf{z}^0] - \mathbb{E} [z_i^l] \mathbb{E} [\mathbf{z}^0]$$

Step 2: Visualize $\text{cov}(z_i^l, \mathbf{z}^0)$ on the board to find *feature detectors*.

In plain English

If a neuron fires whenever there is a bishop on a particular diagonal, its covariance with those input squares will be high — the map reveals *what the neuron is looking at* in the input.

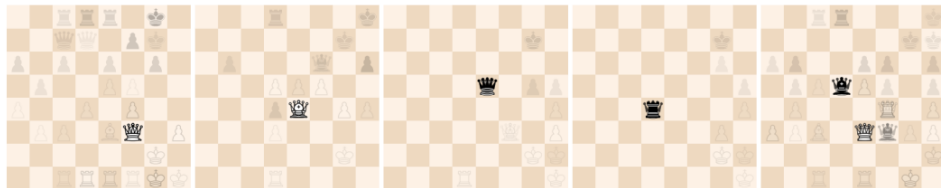
Key Insight

Unlike NMF (which finds group patterns across channels), this method zooms into **individual neurons** — asking *what specific input configuration drives this unit?*

🍷 Results: Input-Activation Covariance Analysis ---

Detecting Move-Types from a Square:

- 1–2) Diagonal-Attacking Pieces (Queen, Bishop)
- 3–4) Horizontally-Attacking Pieces (Queen, Rook)
- 5) Both



5 covariances with different channels for the square (5, 4) E4

Conclusion: Key Findings (Paper 1) ---

- ① Human-Defined Concepts can be regressed from the AlphaZero network
 - ▶ Despite never being trained on a human game of chess
- ② As training progresses, AlphaZero understands basic concepts (openings, material) before more complex ones (king safety, mobility)
- ③ Feature Detectors of human-recognizable chess concepts are encoded by individual layers + channels in AZ Network
 - ▶ Can be found using supervised & unsupervised techniques

Limitations ---

- ① Challenges for Concept Probing (previous section)
- ② Knowledge acquisition is not complete — only a very small part of the model
- ③ Interpreting “Feature Detectors’ ’ found via Unsupervised techniques
 - ▶ Inherently subjective
- ④ No Causal Insight into the Learned Concepts (only correlation)

Future Research Areas ---

- ① Addressing Concept Probing Limitations (more than sparse LR)
- ② Can we go beyond finding human knowledge embedded in AlphaZero and understand what new concepts are learned?
 - ▶ Further analysis of feature detectors found with unsupervised techniques
- ③ How do we generalize these findings to other machine learning settings? Could we find human-recognizable concepts in different trained models?

Discussion (Paper 1) ---

- How is this paper different from methods we discussed so far?
 - ▶ In terms of goals / techniques / specific models to explain
- Does this approach have potential applications for a more general setting other than board games?
 - ▶ What are the pros/cons of using games/artificial environments as baseline for evaluating RL algorithms?
- Can this be applied to our practice of doing science?
 - ▶ Can we recreate or extract theories using similar frameworks (AI for science)?
- How should we define or understand “knowledge” in these settings?
- How convinced are you about the “interpretability” aspect?
 - ▶ Who would be potential audience that could benefit from such analysis?

What the DAAM: Interpreting Stable Diffusion Using Cross Attention

Tang, Liu, Pandey, Jiang, Yang, Kumar, Stenetorp, Lin & Ture (2023)

Presented by: Alex Lin, Jason Jabbour, Mark Mazumder

🍷 Outline ---

- Related work
- Background: Stable Diffusion
- **DAAM** for text-to-image **attribution**
- Results
 - ▶ Attribution quality analysis
 - ▶ Syntax to pixels
 - ▶ Entanglement
- Limitations & discussion

DAAM estimates per-pixel attribution for each word in a prompt (post-hoc)



🍷 Related Work ---

This work applies existing techniques (cross-attention) to an **open source, SOTA** diffusion model to probe limitations

- **Textual perturbation** (Wallace et al., 2019), Attentional Visualization, information bottlenecks to relate important input tokens to outputs of large LMs
- Probing **vision transformers for verb understanding** (Hendricks & Nematzadeh, 2021)
- Enhancing diffusion models using **prompt engineering** (Hertz et al., 2020; Woolf, 2020)
- **Disentangling** e.g., style and spelling (Karras et al., 2019; Materzynska et al., 2022)
- Counterfactual explanations for *text classification* (Jacovi et al., 2021) — unclear how to extend to image generation

🍷 Background: Generative Models ---



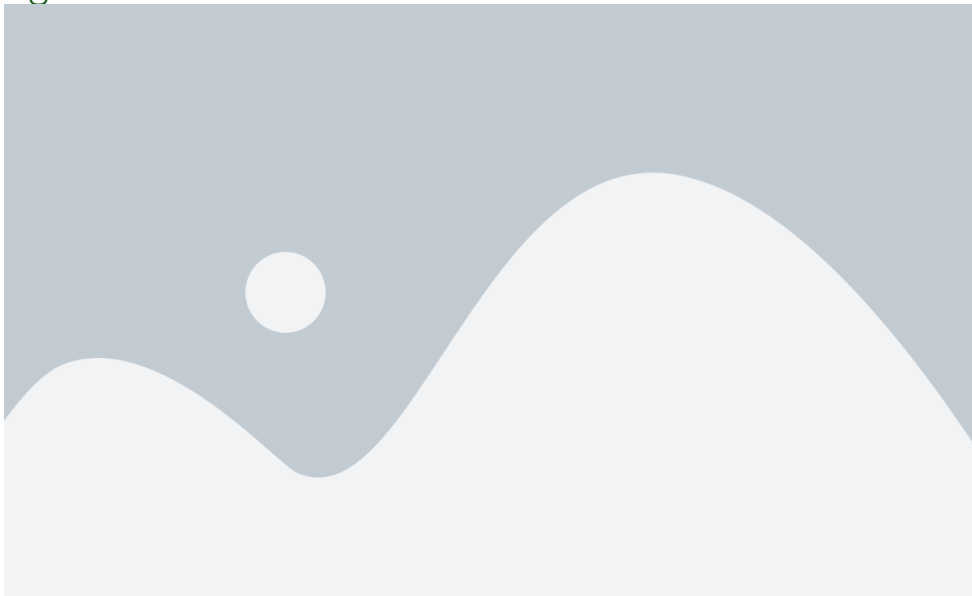
🍷 Background: Image Generation with Diffusion ---



🍷 Background: UNet Architecture ---



🍷 Background: Forward and Reverse Process ---



🍷 Diffusion Model Loss ---

$$L_{DM} = \mathbb{E}_{x, \epsilon \sim \mathcal{N}(0,1), t} \left[\|\epsilon - \epsilon_{\theta}(x_t, t)\|_2^2 \right]$$



🍷 Latent Diffusion Model Loss ---

$$L_{DM} = \mathbb{E}_{x, \epsilon \sim \mathcal{N}(0,1), t} \left[\|\epsilon - \epsilon_{\theta}(x_t, t)\|_2^2 \right]$$

$$L_{LDM} := \mathbb{E}_{\mathcal{E}(x), \epsilon \sim \mathcal{N}(0,1), t} \left[\|\epsilon - \epsilon_{\theta}(z_t, t)\|_2^2 \right]$$

- $\mathcal{E}(x)$: latent space encoder
- z_t : latent representation of image at time t
- Easier to estimate noise in latent space!

(Rombach et al. 2022)

🍷 Stable Diffusion Architecture ---



🍷 Cross-Attention for Text Conditioning ---

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right) \cdot V$$

$$Q = W_Q^{(i)} \cdot \varphi_i(z_t), \quad K = W_K^{(i)} \cdot \tau_\theta(y), \quad V = W_V^{(i)} \cdot \tau_\theta(y)$$

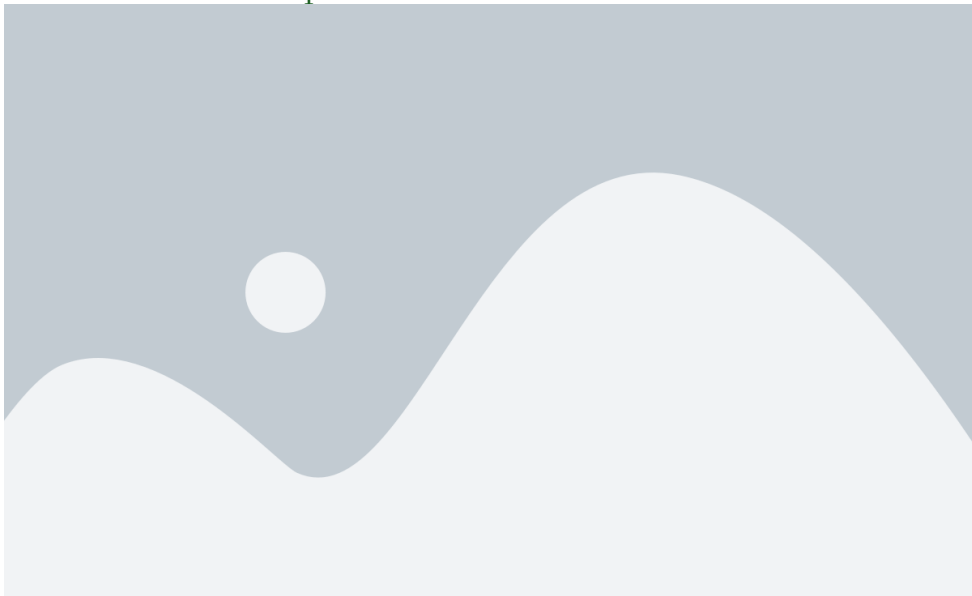
- **Queries:** image tokens (UNet block i)
- **Keys, Values:** prompt tokens (text encoder τ_θ)

DAAM estimates **per-word attribution** via this cross-attention **for each subset** of prompt tokens

🍷 DAAM: Per-Word Heatmaps ---



🍷 DAAM: Attention Maps Formula ---



🍷 Results: Attribution Analysis Part 1 -- Object Attribution ---

We can evaluate DAAM as an image-segmentation tool



🍷 DAAM Segments Stable Diffusion Images ---



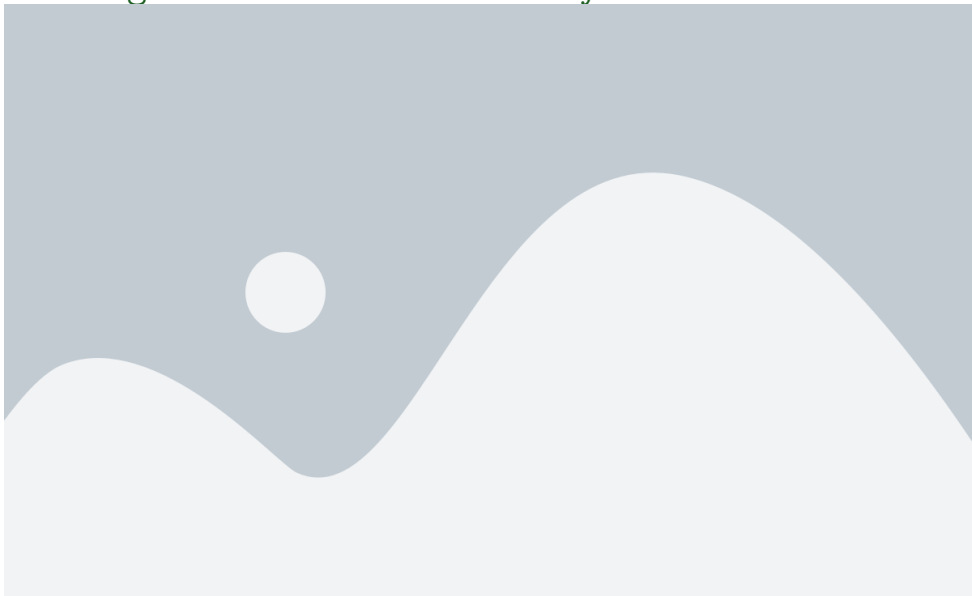
- DAAM does not require explicit segmentation labels
- DAAM is “open vocabulary” — can segment for **any text input** (not limited to known classes)
- Competitive with unsupervised methods; lower than supervised

🍷 Results: Attribution Analysis Part 2 -- Generalized Attribution ---

DAAM can segment **beyond nouns**:



🍷 Evaluating DAAM with a User Study ---



Key Question

How does **syntax** relate to the generated **pixels**?

- Study **head-dependent pairs** in sentences using Universal Dependencies (UD) syntax
- For each dependency relation, measure how much the attribution map of the *dependent* overlaps with that of the *head*
- 3 measures: mIoU, mIoD (Mean IoD over the **D**ependent), mIoH (Mean IoD over the **H**ead)

🍷 Visuosyntactic Analysis: Head-Dependent Pairs ---



🍷 Visuosyntactic Analysis: Measures of Overlap ---



Measures of Overlap

- Mean Intersection over Union (mIoU)
- Mean Intersection over the Dependent (mIoD)
- Mean Intersection over the Head (mIoH)

$mIoD > mIoH$: dependent's map **subsumes** head's

$mIoD < mIoH$: head's map **subsumes** dependent's

🍷 Visuosyntactic Analysis: Results ---



🍷 Visuosyntactic Analysis: Key Takeaways ---

- **Noun Compounds** (“ice cream’ ’): No dominance — complement one another
- **Punctuation & Articles** (“the donut’ ’): No dominance — little semantic meaning
- **Verb & Subject** (*bird stands*): Head verb **dominates** — contextualises subject/object in surroundings (*semi-intuitive*)
- **Nominal Dependents** (“pile of oranges’ ’): Head dominates — *intuitive*
- **Adjective Modifiers** (“wooden bench’ ’): Dependent **dominates** — *counter-intuitive*

Summary

Attribution map of the **dependent subsumes** that of the head, and vice versa for others.
Dominance is intuitive in some cases but **counter-intuitive** in others.

🍷 Visuosemantic Analysis: Cohyponym Entanglement ---

Key Question

Do semantically **similar words** have **worse** generation quality?



Cohyponym: “a giraffe and a zebra”

- SD image generation **worsens**
- Generates **one** noun but **not both**
- Attribution maps **overlap** → Feature Entanglement

Non-Cohyponym: “a zebra and a fridge”

- Generates **both** nouns
- Attribution maps are **distinct**

🍷 Visuosemantic Analysis: Adjectival Entanglement ---

Setup

Prompt: "<adj> <noun> <verb phrase>" (e.g., "a [rusty] shovel sitting in a clean shed")
If **no entanglement**, background should **not gain attributes** pertaining to that adjective



Takeaway

Attribution maps for adjectives attend **too broadly** across images, beyond the nouns they modify → **Feature Entanglement**

Summary ---

- DAAM provides pixel-level attribution maps for Stable Diffusion, a state-of-the-art text-to-image generator
- These maps appear to be informative, as evaluated through segmentation tasks and user study
- DAAM can be a useful tool for further understanding and analyzing Stable Diffusion — e.g. through visuosyntactic analysis and visuosemantic analysis

Class Discussion ---

- Does DAAM give a clear understanding about how a large-scale latent diffusion model synthesizes text to image and which parts of an image are influenced the most?
- Does DAAM explain all the dynamics of how images are synthesized? If not, how should DAAM be modified to better explain image generation?
 - ▶ Other explanatory tools besides **attention** and **segmentation proposals**?
- DAAM pointed out failure cases of Stable Diffusion. Are there further interpretability methods needed to understand why **feature entanglement** is occurring and how it could be improved?

Thank You!



References --- |

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